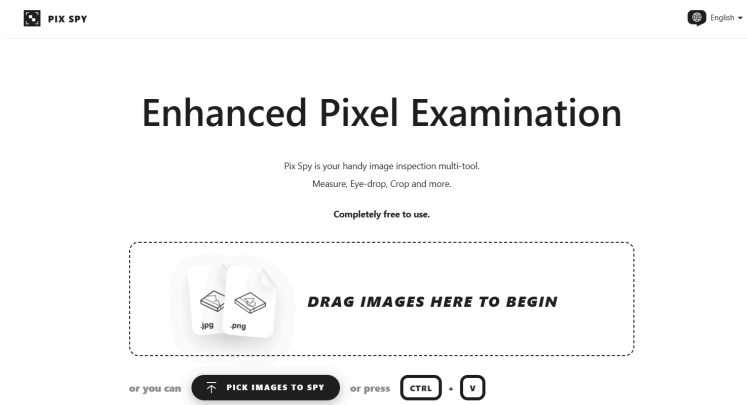
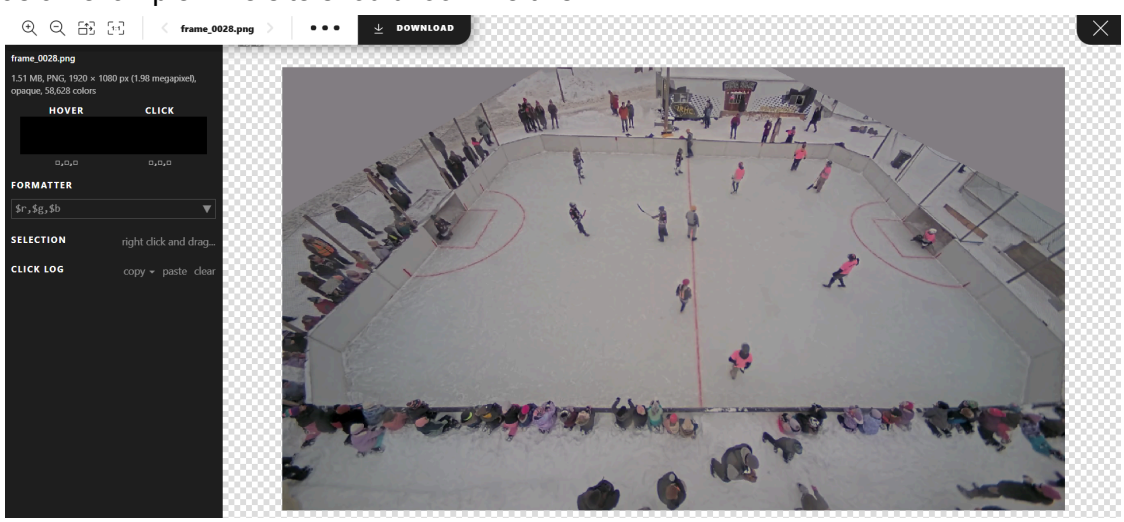
Hi all, this document serves as a tutorial on creating annotations for the images. Let me know if you have any questions! Thank you!

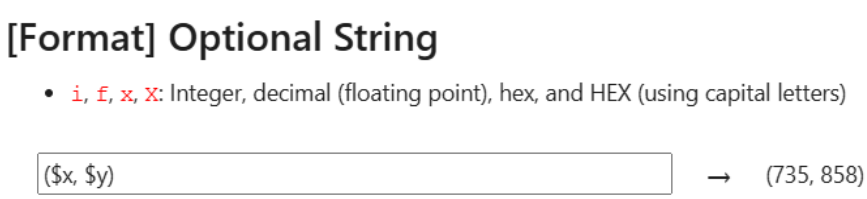
1. Download the masked frames from the shared drive unlabeled_frames. Those frames are extracted from the broomball game on February 10th, 10:10 am - 11:00 am, with a uniform 20 seconds in between frames.
2. Labeling the coordinates of all the people in the image
    a. First, visit the site https://pixspy.com/



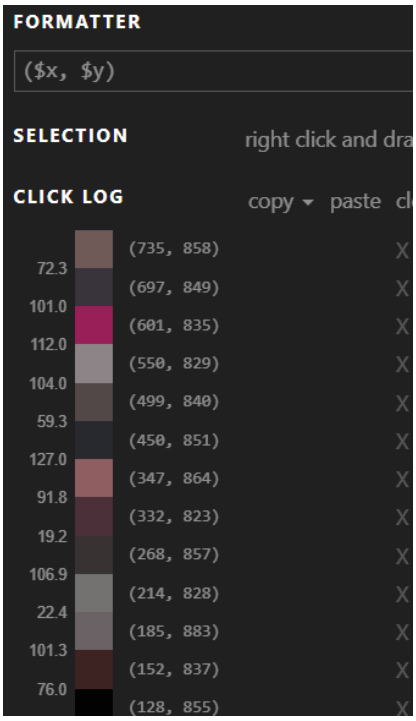
    b. Then, in the download image folder, press (Ctrl + C) to copy the image and then press (Ctrl + V) to paste the image to the site. Using image 28 (frame_0028.png) as an example. The site should look like this:

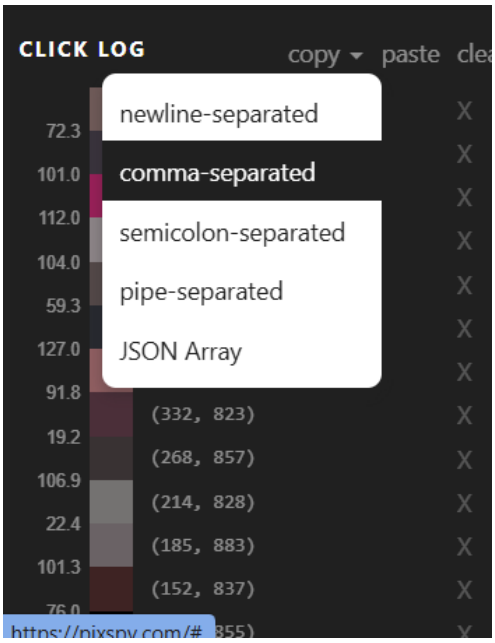


    c. For the formatter option, change the RGB value ($r, $g, $b) format to x and y coordinates ($x, $y) using the “add custom formatter”. Make sure to use parentheses and click add.



d. Now, each click on the image will return a set of xy coordinates. Start clicking each person's head to record the coordinates. You can scroll up and down to zoom in and out of the image. If you click something by accident, click the X next to that coordinate to clear it.





e. Once you have all the coordinates, click on the comma-separated option in the copy; this will let you copy all the coordinates. For example, (272, 299),(238, 235),(210, 252),(289, 384),(259, 397),(69, 418),(76, 467),(57, 504),(174, 507),(89, 571),(37, 835),(86, 851),(128, 855),(152, 837).

3. Compile the coordinates into an npy file of points.
   a. Create a new folder named annotations in your working folder. This is where the code will save the npy file.
   b. Download and open the annotation.npy file from the drive.
   c. Type the ID for the image you are working on and then paste the coordinates you copied from the site. Make sure to add brackets around the points array.

```
# paste the coordinates from (https://pixspy.com/), copy option is "JASON Array"
image_id = 28 #(be sure to change the image id correspondingly)
points = [(272, 299),(238, 235),(210, 252),(289, 384),(259, 397),(69, 418),(76, 4
```

   d. Click run. There should be a check image pop-up showing you the coordinates you saved as the red dots on each person's head. Check and see if there's any mismatch. The run will also automatically save the npy file to the annotations folder.
4. Once you finish, please upload all the .npy files to the annotations folder in the shared drive annotations.